

1. [5 pts] Find the general antiderivative of the following functions. Use C as an arbitrary constants. and use $F(x)$ as the general antiderivative of $f(x)$.

1). $f(x) = 2 \cos x$

$$F(x) = 2 \sin x + C$$

2). $f(x) = \frac{1}{x^3} = x^{-3}$

$$F(x) = -\frac{1}{2} x^{-2} + C \left(= -\frac{1}{2x^2} + C \right)$$

2. [5 pts] Find the following indefinite integrals. Use C as an arbitrary constants.

1). $\int \frac{1}{1+x^2} dx$

$$= \arctan x + C$$

2). $\int \frac{x^4 + \sqrt{x}}{x} dx$

$$= \int (x^3 + x^{-1/2}) dx$$

$$= \frac{1}{4} x^4 + 2x^{1/2} + C$$

$$\left(\text{or } = \frac{1}{4} x^4 + 2\sqrt{x} + C \right)$$